

PROJECT GUIDE BOOK

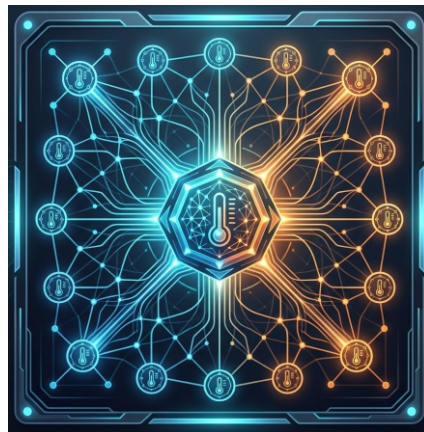
Distributed Smart Temperature Monitoring and Sensor Fusion Network

Second Year Engineering Project

EE2120

Department of Electrical and Electronic Engineering

University of Peradeniya



Academic Year: 2025/2026

Duration: 12 Weeks

Group Size: 6 Students

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1. Introduction

Modern industrial and environmental monitoring systems increasingly rely on distributed sensor networks, wireless communication systems, intelligent data fusion algorithms, and centralized supervision systems. This project introduces students to the practical implementation of an Industrial Internet of Things (IIoT) based smart sensing platform.

Students are required to design and implement a distributed temperature monitoring node using two temperature sensors with different characteristics such as accuracy, response time, noise performance, cost, and linearity. The developed node shall estimate temperature using principles of sensor fusion, transmit the estimated temperature and GPS coordinates to a central SCADA server over WiFi, and collectively contribute to a distributed sensor network capable of estimating temperature at arbitrary spatial locations.

The project integrates knowledge and practical skills related to:

- Embedded systems
- Sensor interfacing
- Signal conditioning
- Wireless communication
- Sensor fusion
- GPS systems
- SCADA systems
- Distributed sensing
- Engineering design methodology
- Team-based engineering practice

2. Learning Outcomes

At the completion of this project, students should be able to:

1. Interface analog and digital sensors with ESP32 microcontrollers.
2. Analyze sensor characteristics such as accuracy, precision, drift, sensitivity, and response time.

3. Implement sensor fusion algorithms for improved estimation accuracy.
4. Develop wireless IoT communication systems using WiFi.
5. Acquire and process GPS coordinate information.
6. Develop communication interfaces between sensor nodes and SCADA systems.
7. Configure and operate SCADA platforms.
8. Implement distributed sensor networks.
9. Estimate environmental variables at unknown spatial coordinates.
10. Conduct experimental validation and performance evaluation.
11. Work effectively in engineering teams.
12. Produce professional engineering documentation.

3. Project Overview

Each student group shall develop one intelligent temperature sensing node.

Each node shall:

Measure temperature using two different sensors.

Fuse sensor outputs to estimate temperature more accurately.

Obtain GPS coordinates using a GPS module.

Transmit data periodically to a central SCADA server over WiFi.

The complete class will form a distributed sensor network. Using the measurements from all groups, students shall:

Create a distributed temperature map.

Estimate temperature at unknown coordinates.

Validate estimation results experimentally.

4. Suggested Hardware Components

4.1. Core Controller

ESP32 Development Board

4.2. Temperature Sensors

Students are encouraged to select two sensors with significantly different characteristics.

Possible sensor options include:

LM35

DS18B20

DHT22

Thermistor

TMP36

PT100/PT1000 with amplifier

Thermocouple module (MAX6675/MAX31855)

4.3. GPS Module

NEO-6M GPS Module

4.4. Communication

ESP32 built-in WiFi

4.5. Optional Components

OLED display

SD card logger

Battery supply

PCB design

Protective enclosure

5. Engineering Tasks

5.1. Task 1 Sensor Characterization

Students shall experimentally determine:

Sensor response time

Sensor noise

Calibration curves

Sensitivity

Accuracy comparison

Students must justify:

Why two sensors were selected

Advantages and disadvantages of each sensor

5.2. Task 2 Sensor Fusion

Students shall implement at least one sensor fusion strategy.

Possible methods include:

Weighted averaging

Adaptive weighted averaging

Moving average fusion

Complementary filtering

Simple Kalman filtering

Confidence-based estimation

Example sensor fusion equation:

$$T_f = w_1T_1 + w_2T_2$$

where:

T_f = fused temperature estimate

T_1, T_2 = sensor measurements

w_1, w_2 = weighting coefficients

with:

$$w_1 + w_2 = 1$$

5.3. Task 3 WiFi Communication

Each node shall:

Connect to WiFi

Transmit:

Timestamp

GPS coordinates

Raw sensor data

Estimated temperature

Node ID

Possible communication protocols:

MQTT

HTTP

TCP Socket

UDP

5.4. Task 4 SCADA Integration

The SCADA platform shall:

Receive data from all nodes

Display real-time temperatures

Show sensor locations

Log historical data

Generate alarms

Suggested SCADA Platforms:

Node-RED Dashboard

Ignition SCADA

ThingsBoard

Grafana + InfluxDB

LabVIEW

5.5. Task 5 Distributed Temperature Estimation

Using all sensor nodes:

Create a distributed temperature map.

Estimate temperature at unknown coordinates.

Possible methods:

Nearest neighbor

Weighted distance interpolation

Inverse Distance Weighting (IDW)

Linear interpolation

Simple machine learning regression

Example estimation equation:

$$T(x, y) = \frac{\sum_{i=1}^n \frac{T_i}{d_i^p}}{\sum_{i=1}^n \frac{1}{d_i^p}}$$

where:

d_i = distance from unknown point to sensor node

p = weighting exponent

6. Project Phases and Timeline

Phase	Duration	Activities	Deliverables
Phase 1	Week 1	Literature review and background study	Literature review report
Phase 2	Week 2	System architecture and design	Design document
Phase 3	Weeks 3-4	Sensor interfacing and calibration	Sensor characterization report
Phase 4	Weeks 5-6	Sensor fusion implementation	Fusion algorithm demonstration
Phase 5	Week 7	GPS and WiFi integration	Functional communication node
Phase 6	Weeks 8-9	SCADA implementation	SCADA dashboard
Phase 7	Week 10	Distributed network implementation	Network demonstration
Phase 8	Week 11	Spatial estimation and validation	Validation results
Phase 9	Week 12	Final presentation and demonstration	Final report and presentation

7. Expected Deliverables

7.1. Technical Deliverables

1. Functional prototype
2. ESP32 firmware
3. Sensor fusion implementation
4. SCADA interface
5. GPS-enabled communication system

7.2. Documentation

1. Design report
2. Calibration report

3. Sensor comparison analysis
4. Mathematical formulation
5. Network architecture
6. Experimental validation
7. User manual

7.3. Presentation

Final demonstration

Viva voce

Poster presentation

8. Suggested Software Tools

8.1. Embedded Development

Arduino IDE

PlatformIO

8.2. Data Analysis

MATLAB

Python

8.3. SCADA and Visualization

Node-RED

Grafana

ThingsBoard

LabVIEW

8.4. Version Control

GitHub

9. Assessment Scheme

Component	Marks
Literature Review	5
System Design	10
Sensor Characterization	10
Sensor Fusion Algorithm	15
Hardware Implementation	15
WiFi and GPS Integration	10
SCADA Integration	10
Distributed Estimation Method	10
Final Demonstration + Poster	10
Report and Presentation	5
Total	100
Bonus Marks for Creativity	20

10. Suggested Advanced Extensions

Students attempting advanced implementations may consider:

Kalman filter-based fusion

Edge AI on ESP32

Adaptive filtering

Fault detection

Cloud integration

Mobile app monitoring

Heat map visualization

Mesh networking

LoRa communication

Energy optimization

Digital twin implementation

11. Experimental Validation

Students should experimentally verify:

- Accuracy improvement due to sensor fusion
- Sensor response during rapid temperature changes
- Communication reliability
- GPS accuracy
- Temperature estimation accuracy at unknown locations

Example experiments:

- Ice-water calibration
- Hot-water transient response
- Outdoor environmental mapping
- Spatial temperature prediction validation

12. Engineering Constraints

Students must consider:

- Cost
- Power consumption
- Scalability
- Communication latency
- Environmental effects
- Sensor placement
- Reliability
- Calibration drift
- Real-time constraints

Role	Responsibilities
Project Leader	Coordination and scheduling
Embedded Engineer	ESP32 programming
Sensor Engineer	Sensor interfacing and calibration
Communication Engineer	WiFi and networking
SCADA Engineer	Dashboard and database management
Data Analytics Engineer	Sensor fusion and estimation

13. Suggested Team Structure

14. Final Expected Outcome

At the completion of the project:

Each group will have developed an industrial-style IoT sensing node.

The class will collectively form a distributed smart sensor network.

Students will gain practical exposure to modern cyber-physical systems and Industry 4.0 concepts.

Applications include:

Smart cities

Industrial plants

Agriculture

Environmental monitoring

HVAC systems

Energy management

Disaster monitoring systems

End of Project Guide Book